

Realization of planar configurations and polytopes

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1 Realization of (planar) configuration

Definition 1.1. Consider a finite set of points $V = \{v_1, \dots, v_n\}$ and lines $E = \{e_1, \dots, e_m\} \subset 2^V$. A configuration is defined as $\mathcal{C} = (V, E)$ where each pair (v_i, e_j) indicates that point v_i lies on the line e_j .

Remark

- What is the definition for a line here? A line here is a subset of V that contain at least two points in V . For example, here is the picture of $V = \{v_1, v_2, v_3\}$ and $E = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_2, v_3\}\}$.

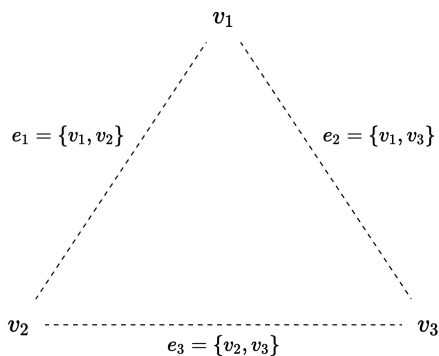


Figure 1: example1

- Like the definition of topological space where we only need a set of points and some open sets. It is an abstract structure, no need to embedded it into other spaces like $\mathbb{R}^n$.
- Through out this paper, we only consider planar configurations.

Definition 1.2. For a configuration $\mathcal{C} = (V, E)$, we define a (affine) realization as a map

$$f : V \longrightarrow \mathbb{K}^2$$

where $f(v_i) \in f(e_j)$ if and only if $v_i \in e_j$ for all $v_i \in V$ and $e_i \in E$. In other words, the coordinate of each point are number in $\mathbb{K}$.

Definition 1.3. Given any field $\mathbb{K}$, we define the projective space $\mathbb{K}\mathbb{P}^n$, also denoted as $\mathbb{P}^n(\mathbb{K})$ or $\mathbb{P}(\mathbb{K}^{n+1})$, as the set of equivalence classes of $\mathbb{K}^{n+1} \setminus \{0\}$ under the equivalence relation $\sim$ defined by $x \sim y$ if there is a nonzero element λ of $\mathbb{K}$ such that $x = \lambda y$. For each element in $\mathbb{K}\mathbb{P}^2$, we can write it as $[x_1, x_2, \dots, x_n]$.

Example: Visualization of $\mathbb{RP}^2$.

- In topology, we know that we can think of $\mathbb{RP}^2$ as gluing each pair of antipodal points of $S^2 \subset \mathbb{R}^3$.
- In differential geometry, I studied that we can even realize $\mathbb{RP}^2$ smoothly as a surface inside $\mathbb{R}^3$, possibly with self-intersections. This can be shown by the famous Boy's surface.

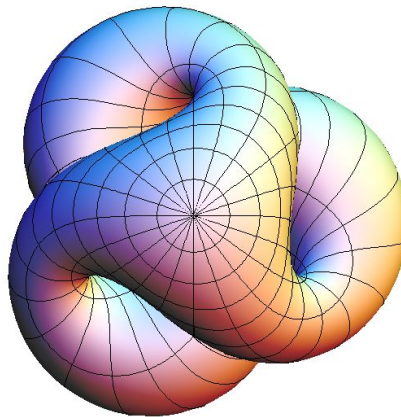


Figure 2: boy's surface in $\mathbb{R}^3$

- For here, I have a better visualization. [3] Pretend that the $w = 1$ plane is the projective plane. There is a pair of parallel lines lying in this plane: $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$. These two lines don't intersect in the ordinary sense, but their intersection in the projective plane is a point at infinity. Let's see what the intersection of their homogeneous counterparts is. Every point on the $w = 1$ plane corresponds to a 3D line through that point and the origin. The two parallel lines thus correspond to two planes. The planes intersect in a line, parallel to the first two, but passing through the origin (and thus lying in the $w=0$ plane). Back in the projective system, this line corresponds to the point at infinity which is the projective intersection of the original two lines.

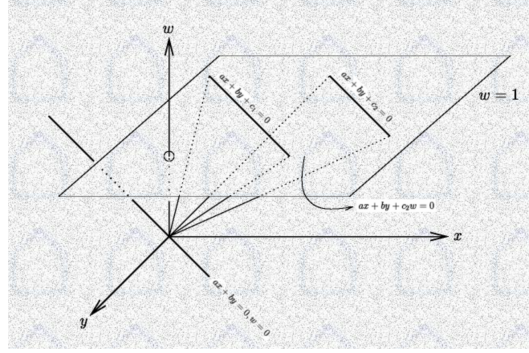


Figure 3: lines in projective plane

In two dimensions, the projective plane is the ordinary real plane augmented by a "line at infinity" r . For each direction d (and $-d$), there is associated a point at infinity $d_\infty = d_{-\infty}$. This point is considered to lie on every line parallel to d . The locus of these points forms the line r .

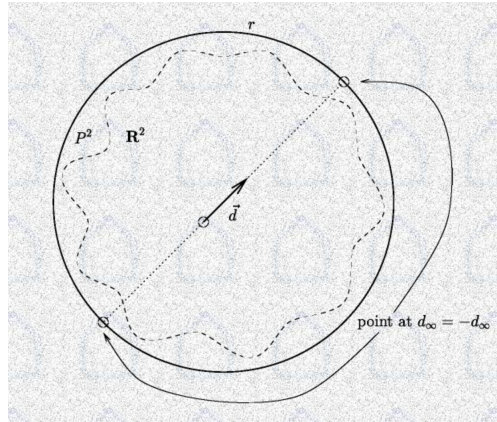


Figure 4: another visualization of $\mathbb{RP}^2$

Definition 1.4. For a configuration $\mathcal{C} = (V, E)$, we define a (projective) realization as a map

$$f : V \longrightarrow \mathbb{KP}^2$$

where $f(v_i) \in f(e_j)$ if and only if $v_i \in e_j$ for all $v_i \in V$ and $e_i \in E$. We should notice that the realization is dependent on the field $\mathbb{K}$.

Remark

- Why we need to use projective space instead of the affine space $\mathbb{K}^2$? Of course we can use this space. For our example1, we define $f(v_1) = (1, 1)$, $f(v_2) = (0, 0)$ and $f(v_3) = (2, 0)$, we

can easily realize it on our Euclidean plane $\mathbb{R}^2$, it becomes a discrete triangle. But if we try to set $f(v_i) = \infty$, here comes the issue. So instead, in order to avoid this issue, we use projective plane. Another good aspect is in projective geometry, all pairs of distinct lines intersect at a unique point, and all pairs of distinct points determine a unique line.

- We can also consider realization of $\mathcal{C}$ over other fields, such as $\mathbb{Q}$, $\mathbb{C}$, or finite fields $\mathbb{F}_q$. Here is a natural question: Is it true that configurations realizable over $\mathbb{C}$ are also realizable over $\mathbb{R}$? What about $\mathbb{R}$ versus $\mathbb{F}_q$? What about different finite fields? The answers to all these questions turns out to be a “NO”.

1.1 Fano configuration

Definition 1.5. Consider this configuration: Given 7 points $V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and lines as $\{\{v_1, v_6, v_2\}, \{v_2, v_4, v_3\}, \{v_3, v_5, v_1\}, \{v_1, v_7, v_4\}, \{v_2, v_7, v_5\}, \{v_3, v_7, v_6\}, \{v_4, v_5, v_6\}\}$. This is called *Fano configuration*.

Lemma 1.1. The projective plane $\mathbb{F}_2\mathbb{P}^2$ is isomorphic to the set of 1-dim subspaces of $\mathbb{F}_2^3$.

Proof. Easy. I will prove this result for any field $\mathbb{K}$. In the definition of $\mathbb{K}\mathbb{P}^2$, each equivalent class is a span of a none zero vector in $\mathbb{K}^3$ without the origin. This correspond to a 1-dim subspace. $\square$

Theorem 1.1. The fano configuration can be (projective) realized over $\mathbb{F}_2$. But it cannot be (affine) realized over $\mathbb{F}_2$.

Proof. Easy, by using the lemma above. We know that

$$\mathbb{F}_2\mathbb{P}^2 = \{(0, 0, 1), (0, 1, 0), (1, 0, 0), (0, 1, 1), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}$$

Recall, for any field $\mathbb{F}$, each element in $\mathbb{F}\mathbb{P}^2$ correspond to a line passing origin in $\mathbb{F}^3$. But what is a line in $\mathbb{F}\mathbb{P}^2$? We defined a line in $\mathbb{F}\mathbb{P}^2$ to be something of the form

$$\{[x_1, x_2, x_3] \in \mathbb{F}\mathbb{P}^2 : \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = 0, \alpha_1, \alpha_2, \alpha_3 \text{ not all zero}\}$$

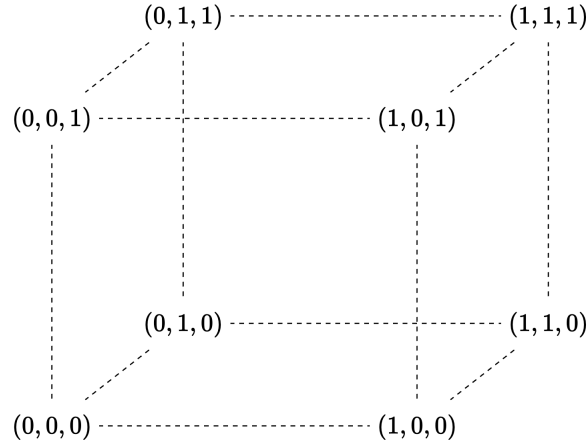


Figure 5: picture of $\mathbb{F}_2^3$

In the case of $\mathbb{F} = \mathbb{F}_2$, $(\alpha_1, \alpha_2, \alpha_3)$ has exactly 7 cases.

$$\begin{aligned}
(\alpha_1, \alpha_2, \alpha_3) &= (1, 1, 1) \text{ so } (x_1, x_2, x_3) = (0, 1, 1), (1, 0, 1), (1, 1, 0) \\
(\alpha_1, \alpha_2, \alpha_3) &= (1, 1, 0) \text{ so } (x_1, x_2, x_3) = (0, 0, 1), (1, 1, 0), (1, 1, 1) \\
(\alpha_1, \alpha_2, \alpha_3) &= (1, 0, 1) \text{ so } (x_1, x_2, x_3) = (0, 1, 0), (1, 0, 1), (1, 1, 1) \\
(\alpha_1, \alpha_2, \alpha_3) &= (0, 1, 1) \text{ so } (x_1, x_2, x_3) = (1, 0, 0), (0, 1, 1), (1, 1, 1) \\
(\alpha_1, \alpha_2, \alpha_3) &= (1, 0, 0) \text{ so } (x_1, x_2, x_3) = (0, 1, 1), (0, 1, 0), (0, 0, 1) \\
(\alpha_1, \alpha_2, \alpha_3) &= (0, 1, 0) \text{ so } (x_1, x_2, x_3) = (0, 0, 1), (1, 0, 0), (1, 0, 1) \\
(\alpha_1, \alpha_2, \alpha_3) &= (0, 0, 1) \text{ so } (x_1, x_2, x_3) = (1, 1, 0), (1, 0, 0), (0, 1, 0)
\end{aligned}$$

This is exactly the fano configuration.

For the second part, just notice $\mathbb{F}_2^2$ has only 4 elements. □

Lemma 1.2 (Gallai–Sylvester theorem). *Let $X \subset \mathbb{R}^2$ be a finite set of points, not all on the same line. Then there exists a line containing exactly two points in X .*

Lemma 1.3. *A projective plane over a field, does contain the Fano configuration if and only if the characteristic of the field is two.*

Proof. Left as homework. □

Theorem 1.2. *The fano configuration cannot be (projective) realized over $\mathbb{R}$. It also cannot be (affine) realized over $\mathbb{R}$.*

Proof. For the second part. Suppose we can. Then by the Gallai–Sylvester theorem above, there exists a line containing exactly two points, this is a contradiction. □

Remark

- In many other books, the fano configuration is shown by the following picture. Fano configuration cannot be realized over $\mathbb{R}$ means there is no way to 'straighten' the middle circle.

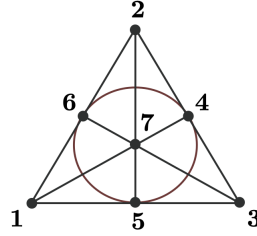


Figure 6: illustration of fano configuration

- Fun fact. We define an automorphism of this configuration to be any permutation of points and lines that preserves the incidence relation. We denote $\mathcal{F}$ to be the fano configuration. Then

$$Aut(\mathcal{F}) \cong GL_3(\mathbb{F}_2)$$

The simple group of order 168. [2]

1.2 Perles configuration

Definition 1.6. Consider 9 points $\{v_1, \dots, v_9\}$ and ten lines below

$$\begin{aligned} &\{\{v_1, v_2, v_3, v_4\}, \{v_1, v_6, v_7\}, \{v_1, v_5, v_9\}, \{v_8, v_5, v_2\}, \{v_8, v_6, v_3\}, \{v_8, v_7, v_4\}, \\ &\{v_9, v_5, v_1\}, \{v_9, v_6, v_2\}, \{v_9, v_7, v_3\}, \{v_4, v_6, v_5\}\} \end{aligned}$$

This system is called Perles configuration, denoted as $\mathcal{P}$.

Theorem 1.3. The perles configuration can be (affine) realized over $\mathbb{R}$. It also can be (projective) realized over $\mathbb{R}$.

Proof. Let us consider the realization $f : V \longrightarrow \mathbb{R}^2$. This can be shown by the following picture.

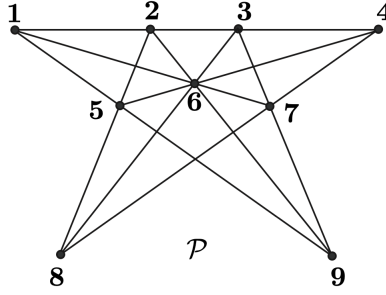


Figure 7: realization of perles configuration on $\mathbb{R}^2$

Now consider the map $\Phi : \mathbb{R}^2 \longrightarrow \mathbb{RP}^2; (x_1, x_2) \longrightarrow [x_1, x_2, 1]$. We can check that by composing this injective map, $\Phi \circ f$ is the realization from V to $\mathbb{RP}^2$. $\square$

Theorem 1.4. The perles configuration cannot be (projective) realized over $\mathbb{Q}$.

Proof. First, we can send the line (1234) to the line at infinity by a projective linear transformation. Recall the definition of projective linear transformation:

$$\begin{aligned} A_{3 \times 3} : \mathbb{KP}^2 &\longrightarrow \mathbb{KP}^2 \\ [x_1, x_2, x_3] &\longrightarrow [x'_1, x'_2, x'_3] \text{ where } \vec{x}' = A\vec{x} \end{aligned}$$

For any line in $\mathbb{KP}^2$, it correspond to a plane $ax + by + cz = 0$ in $\mathbb{R}^3$, we can use a to send this plane to the plane $z = 0$ which correspond to the infinity line in $\mathbb{KP}^2$. By previous observation, after the trasformaton, the line (58), (69) are parallel, the line (78), (5, 6) will be parallel. Now just consider the pentagon [56789], as following, notice also (79), (68) must be parallel

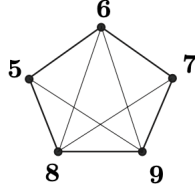


Figure 8: pentagon [56789] after transformation

We then simply do a transformation to make the pentagon becomes following: (here we must assume the pentagon in the picture above has rational coordinates)

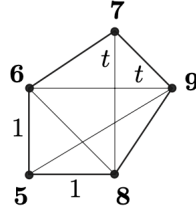


Figure 9: pentagon [56789] after second transformation

By simple geometry, $t = -\frac{\sqrt{5}+1}{2}$ or $t = \frac{\sqrt{5}+1}{2} - 1$. To summarize, if perles configuration has a rational realization, there would be a rational projective linear map into a nonrational arrangement, a contradiction. So we proved that it has no (projective) realization over $\mathbb{Q}$. For nonrealizable (affine) over $\mathbb{Q}$, proof is almost the same. We assume it has a rational realization, and use $\Phi : (x_1, x_2) \longrightarrow [x_1, x_2, 1]$. $\square$

Remark:

- So there is some configuration, that however we realize it on the Euclidean plane, we can never get all rational coordinates!
- In the 1960's, Grunbaum conjectured that Perles' construction is the smallest: any planar configuration on eight or fewer points if it is realizable with real coordinates in the plane, it is also realizable with rational coordinates. Surprisingly, a professor at UBC gave a proof for this conjecture this summer! [4] I am not sure if this is the first proof.
- A natural question arises: What about polygons? If we can stretch it without destroying its topological structure, can we put all vertices in rational coordinates?

2 Realization of polytope

Definition 2.1. The most familiar definition of a polytope $P \subset \mathbb{R}^n$ is define $P := \text{conv}(S)$ where S is a finite set of points. Equivalently, we can define it as the intersection of finitely many halfspaces

but here we require the intersection to be bounded.

Remark:

- Like topological spaces and our point-line configuration before, we want an intrinsic structure.

Definition 2.2. *The face lattice of a polytope is a structured way of organizing and representing all the faces of a polytope in terms of their inclusion relationships. It captures the hierarchical structure of faces in a polytope, from the lowest-dimensional faces (vertices) to the highest (the polytope itself).*

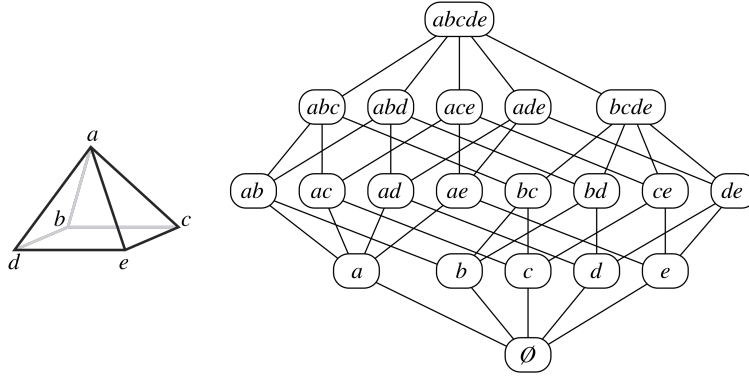


Figure 10: Pyramid face lattice

We say two polytopes are combinatorically equivalent if they have the same face lattice.

Definition 2.3. *We define a realization of a face lattice is to find a polytope in $\mathbb{R}^n$ which has this face lattice. When we say realization of a polytope, we mean realization of its face lattice.*

Theorem 2.1* (Perles, Mnev). *There exists a polytope $P \subset \mathbb{R}^d$ which cannot be realized over $\mathbb{Q}$. Moreover, there exists a polytope $P \subset \mathbb{R}^d$ which cannot be realized over any finite extension of $\mathbb{Q}$.*

3 A glimpse of universality

Following venerable traditions for example from Algebraic Geometry (where one speaks of “moduli spaces”) it is natural and profitable to study not only special realizations for discrete-geometric structures such as configurations, polytopes. We want to study the space of all realizations which is known as the realization space of the structure.

Theorem 3.1* (Grunbaum). *The realization space of any configuration, polytope or polyhedral surface is a semi-algebraic set.*

Remark:

- For what is a semi-algebraic set, why it is a semi-algebraic set, what is realization of polyhedral surface, read the article in the reference.[5]

References

- [1] I. Pak, *Lectures on Discrete and Polyhedral Geometry*, April 20, 2010.
- [2] D. S. Dummit and R. M. Foote, *Abstract Algebra*, Third Edition, University of Vermont.
- [3] *Projective Transformations*, available online at <https://courses.cs.washington.edu/courses/cse557/97wi/notes/xforms/projective.html>.
- [4] J. Solymosi, *On Perles' Configuration*, arXiv preprint arXiv:2408.09370, August 18, 2024.
- [5] G. M. Ziegler, *Non-rational Configurations, Polytopes, and Surfaces*, Inst. Mathematics, MA 6-2, TU Berlin, October 24, 2007; revised November 16, 2007.